Chi-square analysis and logistic regression for CN vs AD classification using ADNI data

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1. Research Objectives

The objective of this study is to analyze whether selected variables are significantly associated with Alzheimer's Disease diagnosis status using ADNIMERGE dataset. The dependent variable is diagnosis status, classified as cognitively normal (CN) or Alzheimer's Disease (AD). Independent variables include gender, age, education, marital status, APOE ε4 status, race and ethnicity. Chi-square tests are used to examine associations between each independent variable and diagnosis, while logistic regression is used to predict AD status.

2. Dataset

This project uses data from Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset (ADNIMERGE subset). The dataset contains demographic, clinical, genetic, cognitive, and imaging-related variables for participants at different stages of Alzheimer's Disease (AD). The dataset was filtered to include only baseline visits and only two diagnosis groups (CN and AD), since study is focusing on binary classification. After filtering the final dataset contained 953 participants, among which 542 CN participants and 411 AD participants.

3. Variables

The dependent variable is diagnosis status:

Variable	**Dataset name**	**Description**
Diagnosis	DX_bl	CN or AD

Table 1. Dependent variables

The independent variables selected were:

Variable	**Dataset name**	**Type**
Age	AGE	Numerical
Gender	PTGENDER	Categorical
Education	PTEDUCAT	Numerical
Marital status	PTMARRY	Categorical
APOE ε4	APOE4	Genetic marker
Ethnicity	PTETHCAT	Categorical
Race	PTRACCAT	Categorical

Table 2. Independent variables

Age and education were grouped only for the Chi-square analysis because the Chi-square test requires categorical variables. In logistic regression, age and education were kept as numerical variables because logistic regression can directly use continuous predictors. Age was grouped into three groups: <70, 70-79 and 80+. Education was grouped similarly to age into three groups: <13, 13-16 and 17+. Race was grouped into white and non-white, while Ethnicity was grouped into Hispanic and Not Hispanic. Race and Ethnicity had to be grouped like this due to lack of diversity in the dataset itself. Unknown values were removed.

4. Methodology

4.1 Chi-square test

The Chi-square test was used to examine whether each selected independent variable had a significant association with diagnosis status. Each variable was separately tested against the diagnosis variable DX_bl.

Hypothesis used for each test:

$H0: There\ is\ no\ significant\ association\ between\ the\ independent\ variable\ and\ diagnosis\ status.$

$H1: There\ is\ a\ significant\ association\ between\ the\ independent\ variable\ and\ diagnosis\ status.$

The significance level was set to:

$$\alpha = 0.05$$

If the p-value was less than 0.05, the null hypothesis was rejected, meaning that variable had a significant association with diagnosis status.

The following Chi-square tests were performed:

1. Gender vs Diagnosis
2. APOE4 vs Diagnosis
3. Marital status vs Diagnosis
4. Age group vs Diagnosis
5. Education group vs Diagnosis
6. Ethnicity vs Diagnosis
7. Race group vs Diagnosis

4.2 Logistic Regression

Since the diagnosis status, consisted of two categories, cognitively normal (CN) and Alzheimer's Disease (AD), binary logistic regression was considered appropriate for this project. For logistic regression, diagnosis status was coded as binary variable, where CN was coded as 0 and AD as 1.

The logistic regression model estimated the probability that a participant belonged to the AD group based on the selected independent variables, presented as following:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Where p represents the probability of being classified as AD, β_0 is the intercept and β_1 to β_n represent the coefficients of the independent variables.

The model coefficients were used to determine how strong relationship was between each independent variable to diagnosis. Odds ratios were calculated from the logistic regression coefficients to make the results easier to interpret. An odds ratio greater than 1 indicated increased odds of being classified as AD, while an odds ratio less than 1 indicated decreased odds of being classified as AD.

5. Results

5.1 Chi square test

Chi-square results are shown in the table below:

Variable	Chi-square value	df	p-value	significant
Gender	7.48	1	0.006239	Yes
APOE4	159.07	2	< 0.0001	Yes
Marital status	35.666	3	< 0.0001	Yes
Age group	19.904	2	< 0.0001	Yes
Education group	46.837	2	< 0.0001	Yes
Ethnicity	0.479	1	0.489	No
Race group	2.028	1	0.1545	No

Table 3. Chi square results

The analysis showed that gender, APOE4, marital status, age group, and education group were significantly associated with diagnosis status. This means that the distribution of these variables differed significantly between CN and AD groups. While, ethnicity and race group were not statistically significant for this classification.

5.2 Logistic Regression Odds Ratio

The first logistic regression results provided odds ratios for the predictors:

Variable	Odds ratio	p-value	interpretation
Age	1.044	0.00042	yes
Male gender	1.379	0.05396	Borderline/no
Education	0.843	<0.0001	Yes
Married	2.476	0.00861	Yes
Never married	1.314	0.60632	no
Widowed	1.16	0.71596	No
APOE4	4.025	<0.0001	Yes
Not Hispanic	1.18	0.7036	No
White	1.25	0.46057	no

Table 4. Logistic regression results

For categorical variables, odds ratios are interpreted relative to a reference category selected by R.

The odds ratio results show that age, education, marital status, and APOE4 were significant predictors of AD status. Age had a positive association with AD, meaning that older participants had higher odds of being classified as AD. Education had a negative association, meaning that participants with more years of education had lower odds of being classified as AD. APOE4 was the strongest predictor in the model. The odds ratio for APOE4 was 4.025, meaning that each additional APOE ε4 allele was associated with approximately four times higher odds of AD diagnosis.

Gender was significant in the Chi-square test, but in logistic regression it became borderline and was not significant at the 0.05 level. Ethnicity and race were not significant predictors in the logistic regression model.

For marital status, the married category had an odds ratio of 2.476, meaning that married participants had approximately 2.48 times higher odds of being classified as AD compared to divorced participants.

5.3 Logistic Regression Likelihood Ratio Test

As categorical variables in logistic regression are displayed through comparison with reference categories, likelihood ratios were used to evaluate each separate predictor.

Variable	df	LRT	p-value	Significance
Age	1	12.695	0.00037	Yes
Gender	1	3.728	0.0535	Borderline/no
Education	1	35.125	<0.0001	Yes
Marital status	3	16.304	0.00098	Yes
APOE4	1	142.478	<0.0001	Yes
Ethnicity	1	0.146	0.70285	No
Race group	1	0.55	0.45844	no

Table 5. Likelihood ratio test

The likelihood ratio test results showed that age, education, marital status, and APOE4 significantly contributed to predicting AD. APOE4 had the strongest contribution, with the largest LRT value and a p-value below 0.001. Gender was borderline but did not reach the standard 0.05 significance level. Ethnicity and race group were again not significant.

The results were consistent, with an exception of gender, which was significant in Chi-square tests but borderline/not significant in Logistic regression test.

5.4 Logistic Regression Model Performance

The logistic regression model was also evaluated using a classification threshold of 0.5. The confusion matrix is shown below:

	Predicted CN	Predicted AD
Actual CN	411	99
Actual AD	146	240

Table 6. Confusion matrix

Metric	Value
Accuracy	72.7%
Sensitivity	62.2%
Specificity	80.6%

Table 7 Model performance

From the table 7 we can see that the model correctly classified 72.7% of participants. 62.2% of AD cases were correctly identified while specificity had better results with 80.6% correctly identified CN participants.

Since the model performance was calculated on the same dataset used to fit the model, these results should be interpreted as an initial performance estimate rather than final generalization performance.

5.5 Additional Chi-square Analysis of Features Used by ML Methods

As an additional analysis, Chi-square tests were performed on the features used by machine learning methods. For kNN, the Tier 1 input features were tested, since kNN does not perform feature selection by itself. For LASSO, the features selected through non-zero coefficients were tested. For Random Forest, the Boruta-confirmed features were used as Random Forest-based selected features.

Since Chi-square tests require categorical variables, continuous imaging and biomarker variables were divided into three groups: low, medium, and high, based on tertiles. Age, education, marital status, ethnicity, and race were tested using the same grouped variables as in the previous Chi-square analysis.

Feature set	Total features tested	significant	Not significant	Significant (%)
kNN Tier 1 input features	17	14	3	82.4%
LASSO selected features	13	10	3	76.9%
Boruta/RF selected features	12	11	1	91.7%

Table 8. Summary of Significance

The results showed that most features used by original machine learning methods were significantly associated with CN vs AD diagnosis. Among the kNN Tier 1 input features, 14 out of 17 features were significant. Among the LASSO selected features, 10 out of 13 features were significant. Among the Boruta/RF selected features, 11 out of 12 features were significant.

Feature set	Non-significant features
kNN Tier 1 input features	PTETHCAT, PTRACCAT, ICV_bl
LASSO selected features	PTRACCAT, ICV, PTETHCAT
Boruta/RF selected features	ICV_bl

Table 9. Non-significant features

The strongest associations were found for imaging and biomarker variables, including hippocampus, entorhinal cortex, FDG, middle temporal volume, APOE4, AV45, ventricles, fusiform, and whole brain volume. In contrast, ethnicity, race group, and intracranial volume were not statistically significant.

Feature	Tested as	Chi-square	p-value
Hippocampus_bl	Tertiles	330.720	<0.0001
Entorhinal_bl	Tertiles	284.753	<0.0001
FDG_bl	Tertiles	275.384	<0.0001
MidTemp_bl	Tertiles	161.439	<0.0001
APOE4	Original categories	159.065	<0.0001
AV45_bl	Tertiles	141.690	<0.0001

Ventricles_bl	Tertiles	120.863	<0.0001
Fusiform_bl	Tertiles	110.861	<0.0001
WholeBrain_bl	tertiles	52.095	<0.0001

6. Discussion

The results show that several demographic, genetic, and imaging-related variables are associated with CN vs AD diagnosis. APOE4 was one of the strongest predictors, being significant in the Chi-square test, logistic regression, and additional ML-feature analysis. Age and education were also significant, where higher age increased the odds of AD, while higher education decreased the odds of AD. Imaging biomarkers such as hippocampus, entorhinal cortex, FDG, middle temporal volume, ventricles, fusiform, and whole brain volume also showed strong associations with diagnosis status.

Ethnicity and race group were not statistically significant in the main Chi-square analysis, logistic regression, or ML-feature analysis. ICV/ICV_bl was also not significant, suggesting that not all variables used in the original machine learning models had a direct statistical association with diagnosis when tested individually.

The difference between Chi-square and logistic regression results for gender is also important. Gender was significant in the Chi-square test but only borderline in logistic regression. This may happen because Chi-square tests one variable at a time, while logistic regression evaluates each variable while controlling for the others.

7. Conclusion

This project analyzed the association between selected variables and CN vs AD diagnosis using the ADNI ADNIMERGE dataset. Chi-square tests showed that gender, APOE4, marital status, age group, and education group were significantly associated with diagnosis, while ethnicity and race group were not significant. Logistic regression showed that age, education, marital status, and APOE4 were significant predictors of AD status. The additional analysis of features used or selected by the original machine learning methods showed that most imaging and biomarker features were also significantly associated with diagnosis. Overall, the results support the relevance of these variables for CN vs AD classification and add statistical explanation to the previous machine learning work.